
Data Structures

BS (CS) _Fall_2024

Lab_11 Tasks



Learning Objectives:

1. Binary Search Tree

Lab Tasks

Submission Instructions

1. Create a separate **.h** and **.cpp** file for each class.
2. Now create a new folder with name *ROLLNO_SEC_LAB11* e.g. **i23XXXX_A_LAB07**(Also name your **cpp** file like this.)
3. Move all of your files to this newly created directory and compress it into a **.zip file**. Return the test file as well.
4. Now you have to submit this zipped file on Google Classroom.
5. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.
6. Plagiarism or AI generated code in the Lab Task will result in **zero** marks in the whole category.

Question 1

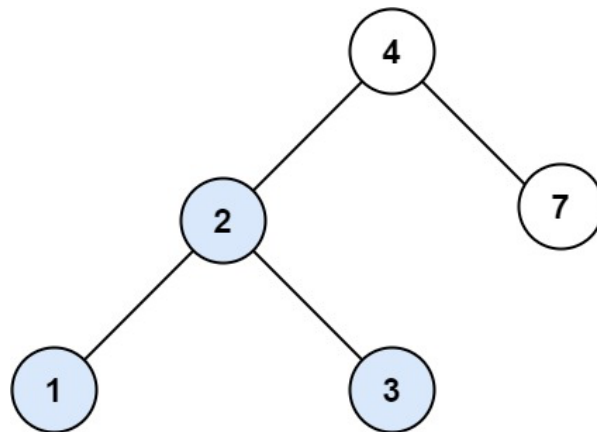
You are given the root of a binary search tree (BST) and an integer val.

Find the node in the BST that the node's value equals val and return the subtree rooted with that node. If such a node does not exist, return null.

Example 1:

Input: root = [4,2,7,1,3], val = 2

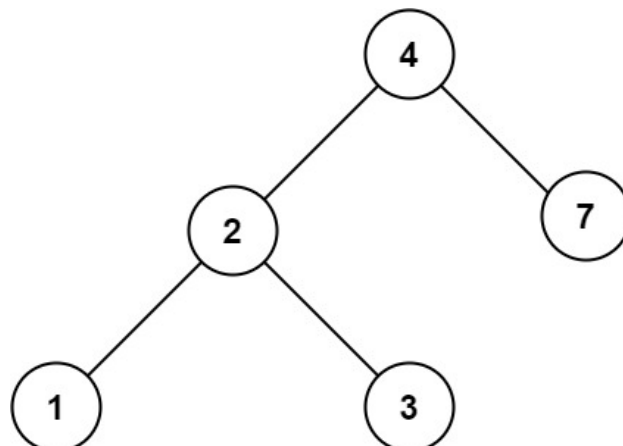
Output: [2,1,3]



Example 2:

Input: root = [4,2,7,1,3], val = 5

Output: []



Constraints:

- The number of nodes in the tree is in the range [1, 5000].
- $1 \leq \text{Node.val} \leq 10^7$
- root is a binary search tree.
- $1 \leq \text{val} \leq 10^7$

Question no. 2

Imagine you are developing a Library Book Management System. In this system, each book has a unique ID, which is an integer. Your task is to create a program that manages these books by using a Binary Search Tree (BST), where each node represents a book with its unique ID.

Requirements:

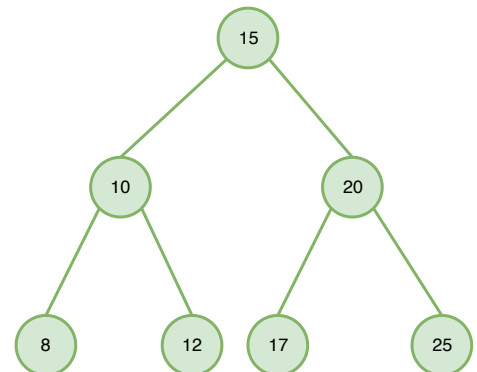
1. **Insert a Book:** Write a function to add a new book by its ID to the BST. Ensure the tree remains a valid binary search tree after each insertion.
2. **Search for a Book:** Write a function to search for a book by its ID. If the book is found, display "Book Found" along with the book's ID. If not, display "Book Not Found".
3. **Delete a Book:** Write a function to remove a book from the tree by its ID. Ensure the tree remains a valid BST after deletion.
4. **Display All Books:** Write a function to display all book IDs in ascending order. This can be achieved by performing an in-order traversal of the BST.
5. **Find the Minimum and Maximum Book IDs:** Write functions to find and display the minimum and maximum book IDs in the library.

Example Interaction:

Enter 1 to Insert a Book
Enter 2 to Search for a Book
Enter 3 to Delete a Book
Enter 4 to Display All Books.
Enter 5 to Find Minimum Book ID
Enter 6 to Find Maximum Book ID
Enter 0 to Exit

For instance, the example uses the following inputs to test the program:

1. Insert books with IDs: 15, 10, 20, 8, 12, 17, 25
2. Search for book ID 10
3. Delete book ID 15
4. Display all book IDs
5. Find the minimum and maximum book IDs



Question no. 3

Given a binary tree, print its nodes level by level in spiral order, i.e., all nodes present at level 1 should be printed first from left to right, followed by nodes of level 2 from right to left, followed by nodes of level 3 from left to right and so on... In other words, odd levels should be printed from left to right, and even levels should be printed from right to left or vice versa.

For example, the spiral level order traversal for the following tree is

(1, 3, 2, 4, 5, 6, 7) or (1, 2, 3, 7, 6, 5, 4)

